INCH POUND
MIL I 7627F
15 November 1994
SUPERSEDING
MIL-1-7627E
4 APRIL 1974

MILITARY SPECIFICATION

# INDICATOR, TURN AND SLIP, 28V DC

This specification is approved for use by all Departments and Agencies of the Department of Defense

## 1 SCOPE

1 1 Scope This specification covers 28V DC, hermetically sealed turn and slip indicators

1 2 Classification The indicators shall be of the following types as specified (see 6 6)

| | |
|---|---|
| MS28041-1 | 4 Minute Turn, Dark Ball, Fluorescent Dial Marking |
| MS28041-1A | 4 Minute Turn, White Ball, Lusterless White Dial Marking |
| MS28041-1B | 4 Minute Turn, Dark Ball, Lusterless White Dial Marking |
| MS28041-1C | 4 Minute Turn, Dark Ball, Fluorescent Dial Marking |
| MS28041-1D | 4 Minute Turn, Dark Ball, Lusterless White Dial Marking |
| MS28041-2 | 2 Minute Turn, Dark Ball, Lusterless White Dial Marking |
| MS28041-2A | 2 Minute Turn, White Ball, Fluorescent Dial Marking |
| MS28024-2 | 4 Minute Turn, White Ball, Lusterless White Dial Marking |
| MS28024-2A | 4 Minute Turn, White Ball, Fluorescent Dial Marking |
| MS28024-3 | 4 Minute Turn, Dark Ball, Fluorescent Dial Marking |
| MS28024-3A | 4 Minute Turn, Dark Ball, Lusterless White Dial Marking |

## 2 APPLICABLE DOCUMENTS

2 1 Government documents

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 6610
DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

2 1.1 Specifications, standards, and handbooks The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6 6)

SPECIFICATIONS

Federal

| | |
|---|---|
| BB-N-411 | Nitrogen, Technical |
| PPP-B-1672 | Boxes, Shipping, Reusables With Cushioning |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Method Of |
| MIL-B-121 | Barrier Material,Greaseproofed, Waterproofed, Flexible |
| MIL-M-3171 | Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion |
| MIL-N-3336 | Nut, Sheet Spring, Instrument Mounting |
| MIL-W-5088 | Wiring, Aerospace Vehicle |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification For |
| MIL-C-5541 | Chemical Conversion Coatings On Aluminum And Aluminum Alloys |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series, General Specification For |
| MIL-A-8625 | Anodic Coatings, For Aluminum And Aluminum Alloys |
| MIL-P-23408 | Plating, Tin-Cadmium (Electrodeposited) |
| MIL-B-81793 | Bearing, Ball, Annular, For Instrument And Precision Rotating Components |
| MIL-C-83488 | Coating, Aluminum ION Vapor Deposited |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-101 | Test Procedures For Packaging Materials |
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL STD 129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of U S Military Property |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-STD-461 | Requirements for the control of Electromagnetic Interference Emissions And Susceptibility |
| MIL-STD-462 | Measurement Of Electromagnetic Interference Characteristics, |
| MIL-STD-704 | Characteristics of Aircraft Electric Power |
| MIL-STD-781 | Reliability Testing for Engineering Development, Qualification & Production |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-831 | Test Reports, Preparation Of |
| MIL-STD-889 | Dissimilar Metals |
| MIL-STD-965 | Parts Control Program |
| DOD-STD-1866 | Soldering Process General (Non-Electrical) |
| MIL-STD-2073-1 | DOD Material Procedures For Development & Application Of Packaging Requirements |
| MS28024 | Indicators, Turn And Slip, 4-Minute Turn |
| MS28041 | Indicator, Turn And Slip, 28V DC, 1-7/8 Inch Dial |
| MS33558 | Numerals And Letters, Aircraft Instrument Dial, Standard Form Of |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094 )

2 2 Order of precedence In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3 1 First article inspection This specification makes provisions for the first article inspection

3.2 Parts and materials In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration In so doing, the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

(b) Non-repairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable The general size of subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity Non-repairable subassemblies must be reliable (see 6 4)

(c) Bearing selection will conform to MIL-B-81793



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3 2 1 Nonstandard parts and materials approval Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-STD-965

3 2 1 2 Recovered materials. Recovered materials is used to the maximum extent possible without jeopardizing the end use of the item

3 3 Design and construction The indicator shall conform with all applicable requirements of MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein

3 3 1 Total weight The total weight of the indicator, less mounting screws, shall be a minimum consistent with good design and shall not exceed 2 0 pounds

3 3.2 Reliability

3 3 2 1 Operational stability The indicator shall operate with satisfactory performance, continuously or intermittently for a period of at least 2500 hours without the necessity for any adjustment

3 3 2 2 Operating life The indicator shall have a minimum total operating life of 25,000 hours with reasonable servicing and replacement of parts Parts requiring scheduled replacement due to wear during the life of the indicator, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity

3 3 2 3 Reliability in Mean-Time-Between-Failures (MTBF) For the test program under 4.6 27, the specified mean operating time between failures shall be 2000 hours For this test program, a failure shall be considered to have occurred whenever the performance characteristics fall below the acceptance requirements (4.6 27.5) irrespective of whether or not the condition can be corrected by adjustment of operator controls

3 3 3 Cabling and connections

3 3 3 1 Cables and connectors The indicator shall provide for the use of cables and connectors in accordance with MIL-E-5400

3 3 3 2 Interconnection cabling The indicator shall be capable of satisfactory operation using external wiring in accordance with the application requirements of MIL-W-5088 The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished External cables and that portion of the connectors attached to the cables shall not be supplied as part of the indicator

3 3 4 Interchangeability The indicator shall meet the interchangeability requirements as defined in MIL-E-5400



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3.3.5 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 measured per MIL-STD-462.

3.3.6 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nameplate. A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with MIL-STD-130, except that the National Stock Number marking shall be omitted.

3.3.8 Standard conditions. The following conditions, unless otherwise specified, shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient (25°± 5° C) |
| Altitude | Normal Ground |
| Vibration | None |
| Humidity | Room ambient up to 90 percent relative humidity |
| Input power voltage | 27.5 ± 0.5V DC |

3.3.9 Environmental conditions. The indicator shall be capable of withstanding the following environmental conditions as required in Section IV herein:

(a) Operating Temperature. Continuous operation over a temperature range of -54° C to 71° C.
(b) Exposure Temperature. Exposure temperature from -62°C to 95° C.
(c) Temperature-altitude. 0 to 70,000 feet.
(d) Temperature-Shock per MIL-STD-810, Method 503.1.
(e) Salt Fog. Exposure to a salt laden atmosphere.
(f) Fungus. Fungus growth as encountered in a tropical climate.
(g) Acceleration. MIL-STD-810, Method 513.2, Procedure I.
(h) Vibration. MIL-STD-810, Method 514.2, Equipment Category b.1, Curve B from 5 to 500 Hz then down the 500 Hz line to the horizontal axis, Procedure I.

3.3.10 Primary input power requirements. The indicator shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from a 28V DC, 0.25 amp power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

3.3.11 Transportability. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibration, and other conditions incident to shipment, storage, installation and service use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3.4 Performance. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail requirements.

3.5.1 Maintenance. The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using tools and items of maintenance equipment which are normally available as commercial standards.

3.5.2 Case. The case shall conform to MS28024 for the 2.75 inch dial indicator and to MS28041 for the 1.87 inch dial indicator except the length may be 4.92 inches. The case and mounting flange shall be made of nonferrous, low density metal, uniform in texture, having a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of durable type to withstand usage encountered in service.

3.5.2.1 Hermetic sealing. The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.5.2.2 Filling medium. The filling medium shall be a mixture of 90 percent nitrogen and 10 percent Helium. The nitrogen used shall be in accordance with BB-N-411, type I, class I, grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ± 0.1 atmosphere.

3.5.2.3 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator.

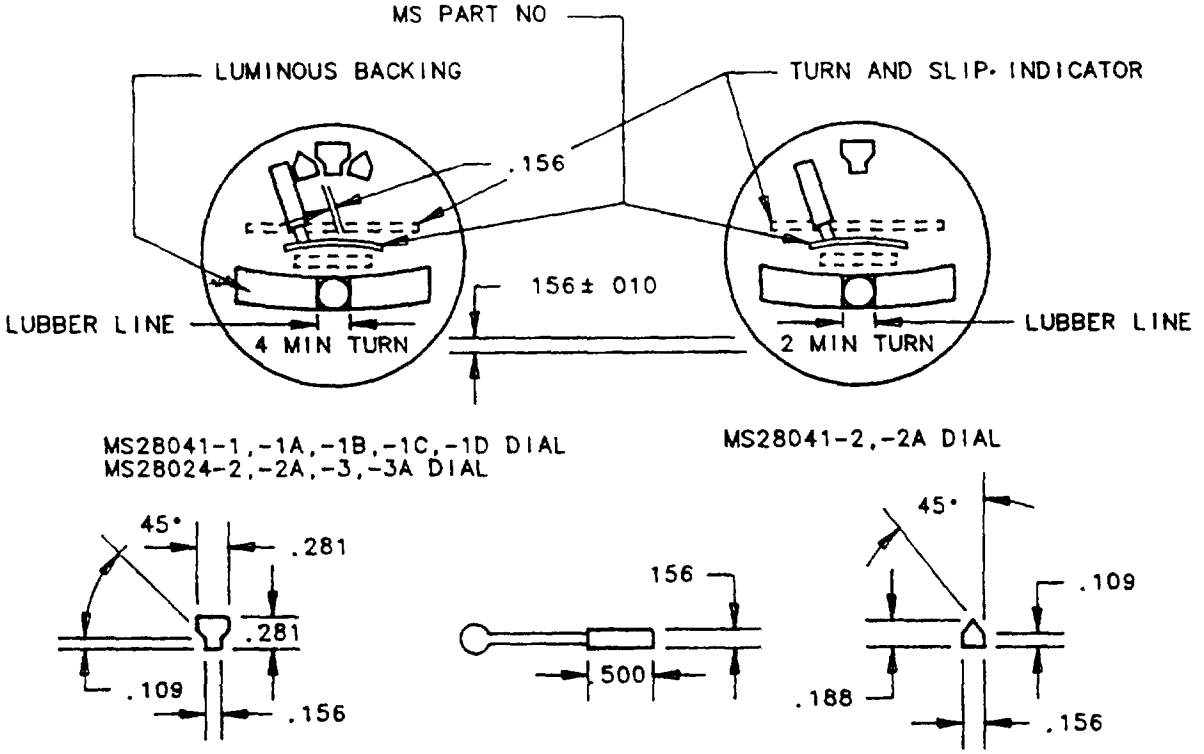
3.5.2.4 Cover glass. The cover glass shall be clear, flat, and free from flaws which interfere with the normal reading of the instrument. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicrons region.

3.5.2.4.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be held to a minimum and shall not exceed 0.188 inch.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

| LUBBER LINE SPACING | |
|---|---|
| 2 INCH INDICATOR | 406 ± 031 |
| 3 INCH INDICATOR | 468 ± 031 |




| MARKING | HEIGHT OR LENGTH | WIDTH OF LINE OR GRADUATION ±.005 | MATERIAL OR FINISH |
|---|---|---|---|
| CENTER MARK | AS SHOWN ON FIGURE | AS SHOWN ON FIGURE | FLUORESCENT LUMINESCENT OR LUSTER-LESS WHITE COLOR NO. 37875 OF FED-STD-595 |
| TWO NEEDLE-WIDTH DEFLECTION MARKS | | | |
| SHADED PORTION OF POINTER | | | |
| REAR HALF AND BOTTOM OF INDICATOR TUBE FOR DARK BALL | | | |
| VISIBLE PART OF TWO WIRES ON FRONT AND TOP OF INCLINOMETER | | | |
| ONE WIRE ON EACH SIDE OF INCLINO-METER BALL IN ZERO POSITION | | | |
| LETTERING, 4-MIN TURN, 2 MIN TURN | | .016 | |
| LETTERING, TURN AND SLIP INDICATOR | .047 | | LUSTERLESS BLACK COLOR NO. 37038 OF FED-STD-595 |
| MS PART NUMBER | | | |
| DIAL BACKGROUND | | | |
| REAR HALF AND BOTTOM OF INDICATOR TUBE FOR WHITE BALL | | | |

Figure 1 Markings, dial and pointer



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3.5.3 Dial. The dial shall conform to figure 1. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to MS33558.

3.5.3.1 Visibility of dial. The pointed, center mark and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.5.4 Slip indicator.

3.5.4.1 Indicator tube glass. The glass of the slip indicator tube shall be made of clear annealed glass tubing free from any flaws that will seriously affect the readability of the inclinometer. The wall thickness shall be not less than 0.05 inch. The inside of the tube shall be smooth and uniform in order that the ball will roll smoothly when the indicator is slowly tipped to either side of the vertical plane of the dial while being tapped gently. The tube shall not leak when held at 71° C for one hour.

3.5.4.2 Indicator ball. When slip indicators with dark balls are specified, the ball shall be of black glass, Color No. 17038, or dark blue glass, Color No. 15044, in accordance with FED-STD-595. When slip indicators with white balls are specified, the ball shall be of white glass. The white color shall be in accordance with the following color notation following Munsell System color notations. The color shall be neutral not less than 9.0 value and no greater than 0.5 chrome. The ball shall be a highly polished accurate sphere and shall be 0.37 ± 0.06 inch in diameter for the 1.87 inch dial indicator or 0.44 ± 0.01 inch in diameter for the 2.75 inch dial indicator.

3.5.4.3 Damping liquid. The damping liquid used in the slip indicators shall be sufficiently colorless to preclude interference with visibility of the ball under daylight and night lighting conditions and under normal temperature and temperature extremes. Damping shall be such that when the indicator is inclined to the horizontal at an angle of 24 degrees, the time for the ball to roll from the zero mark to the end of the tube shall be not less than 0.2 second.

3.5.4.4 Slip indicator zero position and sensitivity. When the indicator is in the normal horizontal position, the ball shall rest at the zero mark within 0.03 inch. With the dial vertical, when the indicator is rotated about the longitude axis until the ball is just short of its limit, the angle of rotation shall be 8 ± 2 degrees for the MS28041 indicator and 10 ± 2 degrees for the MS28024 indicator. The ball shall not stick at the high end of the tube.

3.5.5 Turn indicator.

3.5.5.1 Sensitivity element. The sensitive element of the turn indicators shall be operated on 28 V DC. The element shall be so mounted that it reacts



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

only to turns about the vertical axis. The indication shall be proportional to the rate of turn about the vertical axis.

3.5.5.2 Pointer. The pointer shall be as shown on figure 1. It shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. It shall be firmly attached to the mechanism and yet be readily adjustable. During static or dynamic balance, the deviation of the pointer from zero shall not exceed 0.010 inch for any stationary position.

3.5.5.3 Mechanism adjustment. Means shall be provided to adjust or correct the sensitivity of the indications and, if necessary, the damping of the indications. The adjusting means shall be set in the neutral position, within an amount of one-quarter part of the total range of adjustment, in order that a sufficiently large range of adjustment in each direction will be available. This means of adjustment shall be reasonably simple to manipulate and of a nature that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.5.6 Screw threads. Screw threads 0.060 inch in diameter or larger shall be in accordance with MIL-S-7742.

3.5.7 Electrical requirements.

3.5.7.1 Insulation. The complete electrical system shall be so insulated from the indicator cases that they will be capable of withstanding a voltage of 500 V DC for a period of 1 minute.

3.5.7.2 Case markings. The following markings, in durable dull white or natural metal finish, shall be permanently and legibly marked on the back of the indicator cases beside the electrical receptacle to identify the electrical connector pins with which they shall be radially aligned:

| Pin | Markings |
|---|---|
| A | +28V DC Power |
| B | Power Return |

3.5.7.3 Electric Connector. The electric connector shall be a sealed connector and shall be as shown on MS28041 or MS28024.

3.5.7.4 Voltage variation. The indicators shall operate satisfactorily with a power source ranging from 26 to 30V DC. The indicator motor shall start and come up to a useful speed with 23 volts applied.

3.5.7.5 Wiring. Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. The wiring shall be insulated from the instrument case.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3.5.8.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.5.8.3 Metals. Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt fog or atmospheric conditions to which the instrument may be subjected when in storage or during normal service life.

3.5.8.3.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

3.5.8.3.2 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. Where abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.5.8.3.3 Aluminum alloy parts. When practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.5.8.3.4 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type plating shall conform to 3.5.8.4.

3.5.8.4 Protective treatment. Where materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.5.8.5 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment shall not be necessary.

3.5.8.6 Fumes and vapors. Materials used in the construction of the indicator shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

3.5.9 Mounting hardware. The contractor shall furnish removable spring nuts conforming to MIL-N-3336 in sufficient quantity to install the instrument. The contractor shall furnish sufficient mounting hardware to install the indicator. The screws shall be round head brass machine screws having a durable, lusterless black, oxidized or nickel finish. The length shall be sufficient to mount the instrument on panels up to 3/8 inch thick.

3.5.9.1 Envelope. An envelope furnished by the contractor shall be packaged with each indicator, shall contain the mounting hardware, and shall be marked as follows:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING HARDWARE

3.5.10 Workmanship. The indicator, including all parts and accessories shall be so constructed and finished that it will be free from all defects which may affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, plating, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.5.10.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. When dimensions and tolerances affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.5.10.2 Fabrication. Machining, drilling, and forming shall be done with the use of accurate templates, jigs or gages.

3.5.10.3 Screw assemblies. Assembly screws shall be tight. The word tight means that the screws cannot be tightened appreciably without damage or injury to the screw threads or heads.

3.5.10.4 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be as free as possible from backlash.

3.5.10.5 Cleanup. The indicator shall be thoroughly cleaned for loose, spattered or excess solder, metal chips and other foreign matter after final assembly. Burrs and sharp edges, as well as rosin flash which might crumble, shall be removed.

3.5.10.6 Soldering. Soldering shall be performed in accordance with DOD-STD-1866.

3.5.11 Magnetic property. When held in various positions around a free magnet at a distance of 5.5 inches in a magnetic field having a horizontal intensity



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

of 0.18 ± 0.01 oersted, the deflection of the magnet shall not exceed 5 degrees with and without power applied.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests)as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First Article Inspection (see 4.4)
(b) Quality Conformance Inspection (see 4.5).

4.3 Inspection conditions. The inspection conditions are described under the individual tests to which they apply.

4.4 First article inspection.

4.4.1 Test sample. Unless otherwise specified, the test samples shall consist of three indicators. The samples shall be identical to the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. For First Article Reliability Tests the sample size shall be a minimum of six and a maximum of nine.

4.4.2 First article tests. The test samples shall be subjected to all the tests described under paragraph 4.6, Test Methods.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

4.4.3 Government tests. When specified by the procuring activity, the samples that were subjected by the contractor to first article tests and an untested sample shall be delivered at the contractor's expense to a specified government laboratory for additional testing (see 6.6.1).

4.4.3.1 Data to accompany test samples. When it required that test samples be furnished to the procuring activity, it shall be accompanied by the following data:

(a) Brief operating data to enable test personnel to correctly operate the equipment.

(b) Engineering data in the form of assembly drawings (2 sets) and calibration tests of the indicators.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of:

(a) Individual Tests (see 4.5.1)
(b) Sampling Tests (see 4.5.2, 4.5.3, 4.5.4)

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6.

(a) Examination of product
(b) Low voltage operation
(c) Manufacturing run-in test
(d) Turn indicator static balance
(e) Turn indicator dynamic balance
(f) Turn indicator sensitivity
(g) Turn indicator damping
(h) Slip indicator visibility
(i) Slip indicator filling
(j) Slip indicator zero position
(k) Slip indicator friction
(l) Slip indicator damping
(m) Slip indicator sensitivity
(n) Seal

4.5.2 Sampling Plan A. Indicators shall be selected at random in accordance with the following schedule and subjected to the following tests as described under 4.6:

| PRODUCTION QUANTITY | TOTAL NUMBER OF SAMPLES |
|---|---|
| 1-10 | 1 |
| 11-60 | 2 |
| 61-135 | 3 |
| 136-235 | 4 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

Each additional 200 or fraction thereof: 1

(a) Individual tests
(b) Magnetic effect
(c) Current
(d) Voltage variation
(e) Slip indicator leak
(f) Transient Susceptability

4.5.3 Sampling Plan B. Three indicators selected at random from the first 20 on contract or order shall be subjected to the following tests as described under 4.6:

(a) Individual tests
(b) Vibration resonance
(c) Low temperature operation
(d) High temperature operation
(e) Temperature-altitude
(f) Low temperature exposure
(g) High temperature exposure
(h) Acceleration
(i) Vibration
(j) Salt fog
(K) Mounting lug pull
(l) Temperature shock
(m) EMI
(n) Helium test
(o) Internal examination

4.5.4 Sampling Plan C. A minimum of 6 and a maximum of 9 indicators shall be selected at random from the first 20 on contract or order and shall be subjected to the following tests as described under 4.6:

(a) Individual tests
(b) Reliability

4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be examined carefully to determine that the material and workmanship requirements have been met. This examination must be performed prior to hermetic sealing and must be verified by a government inspector.

4.6.2 Low voltage operation. The electrical power, with the voltage reduced to 23 volts, shall be applied to the indicator. The indicator motor shall start and come up to a useful speed.

4.6.3 Manufacturing run-in test. Each indicator shall be operated under the conditions specified herein for a period of 6 hours without failure. A



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

failure shall be defined as anything which causes malfunctioning of the indicator. Only those adjustments will be permitted which can be made by using such during the normal use of the indicator.

Temperature, ambient room
Humidity, ambient room
Vibration: Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3g's

The indicator shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the indicator shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the indicator in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the indicator shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failures occur, corrective action shall be taken to eliminate this defect from future indicators. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.6.4 Turn indicator static balance. The indicator shall be placed in a stationary position and the pointer checked with no power applied. The deviation of the pointer from zero shall not exceed 0.010 inch for any stationary position.

4.6.5 Turn indicator dynamic balance. The indicator shall be placed in the normal stationary position and the pointer position checked with the gyro operated on 28V DC. The deviation of the pointer from zero shall not exceed 0.010 inch for any stationary position.

4.6.6 Turn indicator sensitivity. The indicator shall be placed in the normal position and operated on 28V DC. The instrument shall be turned about the vertical axis at the rates specified below. The deflection of the pointer also shall conform to the tolerances specified in Table I. This test shall be made after the power has been applied for a period of not more than 3 minutes. The rates of turning may be measured with a timer, or one indicator may be accurately checked against a timer and then used as a standard for testing other indicators. The movement of the pointer shall be smooth throughout its range.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

TABLE I Pointer Deflection

| Rate of turning in degrees per minute | Deflection of pointer tips in inches | |
|---|---|---|
| | MS28041-1, -1A, -1B, -1C, -1D MS28024-2, -2A, -3, -3A | MS28041-2, -2A |
| 36 | 0 06 ± 0 01 | - - |
| 90 | 0 16 ± 0 03 | 0 08 ± 0 03 |
| 180 | 0 31 ± 0 06 | 0 16 ± 0 03 |
| 360 | 0 63 ± 0 06 | 0 31 ± 0 06 |

4 6 7 Turn indicator damping The indicator shall be placed in the normal operating position on a rate table and operated for 5 minutes The indicator shall then be rotated about its vertical axis at a rate which causes full scale deflection of the pointer and then the rotation shall be stopped suddenly The motion of the pointer shall be so damped that the pointer returns to the zero mark without crossing it in not less than one second, and not more than 3 seconds, and without oscillation or noise other than that encountered in normal operation

4 6 8 Slip indicator visibility. The indicator shall be so tipped that the ball is at rest at either end of the tube. No less than one-half of the ball shall be visible when the ball is viewed from a position 12 inches directly in front of the zero mark of the slip indicator

4 6 9 Slip indicator filling The indicator shall be so tipped that all the air in the tube is trapped in the expansion chamber end of the tube. The position of the mounting lugs of the indicator shall be in a vertical plane with the line joining the centers of the two lower mounting holes in a horizontal plane. No part of the air bubble shall be visible when the indicator is viewed from a position 12 inches directly in front of the zero mark of the slip indicator.

4 6 10 Slip indicator zero position The two lower adjoining mounting holes of the indicator shall be placed in the normal horizontal position, and the indicator shall be gently tapped The glass ball shall rest at the zero mark of the slip indicator within 0 03 inch

4.6 11 Slip indicator friction. The indicator shall be slowly tipped to either side of the vertical plane of the dial The ball shall roll smoothly when the indicator is gently tapped

4 6 12 Slip indicator damping The plane of the mounting lugs of the indicator shall be placed in a vertical position with the line joining the centers of the two lower or two upper mounting holes inclined to the



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

horizontal at an angle of 24 degrees, in order that the ball will roll to the opposite end of the tube. The time of roll of the ball from the zero mark of the indicator to the end of the tube shall be not less than 0.2 second.

4.6.13 Slip indicator sensitivity. With the dial vertical, the indicator shall be rotated about the longitudinal axis to the right until the ball is just short of its limit. The angle of rotation shall be 8 ± 2 degrees for MS28041 and 10 ± 2 degrees for MS28024. The test shall be repeated rotating the instrument to the left. The same tolerances shall apply. The ball shall not stick at the high end of the tube.

4.6.14 Seal. The indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector and the differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately 1 atmosphere. The leak rate shall be based on a 1-atmosphere differential pressure with pure helium as the filling medium. Under these conditions, (1 atmosphere differential pressure with pure helium as the filling medium) the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer. The leak test shall be conducted either with the instrument evacuated and surrounded by helium, or after instrument has been filled. As an alternate test, an immersion leak test may be performed in accordance with Procedure 1, Method 512.1, of MIL-STD-810.

4.6.15 Magnetic effect. The indicator shall be tested with no power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5.5 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity if between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in position 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed 5 degrees. This test shall be repeated with power applied to the indicator. The same tolerance shall apply.

4.6.16 Vibration. The indicator shall be tested in accordance with Method 514.1, Equipment Category A, Curve B, Procedure I of MIL-STD-810. Throughout this frequency range, the maximum magnitude of pointer vibration shall not exceed 0.03 inch double amplitude, and the slip indicator ball shall not differ from its zero position by more than 0.06 inch.

4.6.16.1 Oscillation. The indicator shall be tested in accordance with MIL-STD-810, Method 514.2, Equipment Category 6.1, Curve B, from 5 to 500 Hz then down the 500 Hz line to the horizontal axis, Procedure I. The peak-to-peak amplitude of oscillation shall be .008 radians with a frequency range of 4 to



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

7 Hz. Under these conditions, the vibration of the pointer shall not exceed .020 inches single or .040 inches double amplitude.

4.6.17 Low temperature operation. (This test may be combined with the low temperature exposure test at the discretion of the testing agency.) The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subject to an ambient temperature of -54°C (-65°F) for a period of 4 hours. At the end of the 4-hour period and with the temperature maintained at -54°C (-65°F), the indicator shall be subjected to and shall meet the requirements specified in the turn indicator sensitivity (low temperature) test and the slip indicator damping (low temperature) test. There shall be no slip indicator leakage as a result of these tests.

4.6.18 Turn indicator sensitivity (low temperature). The indicator shall be mounted in the normal operating position on a rate table and operated for 10 minutes maximum. The indicator shall then be rotated about the vertical axis at 180 degrees per minute for MS28041-1, MS28041-1A, MS28041-1B, MS28041-1C, MS28041-1D, MS28024-2, MS28024-2A, MS28024-3, MS28024-3A, and 360 degrees per minute for MS28041-2 and MS28041-2A. The pointer deflection shall be 0.31 ± 0.06 inch. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 0 ± 0.01 inch. The test shall be performed in both directions of rotation.

4.6.19 Slip indicator damping (low temperature). The slip indicator damping test shall be repeated and the time for the ball to travel from the zero mark to the extreme end of the tube shall not be greater than 4 seconds.

4.6.20 High temperature operation. (This test may be combined with the high temperature exposure test at the discretion o the testing agency.) The indicator shall be properly connected and power shall be applied. The indicator shall then be subjected to an ambient temperature of 71°C (160°F) for a period of 4 hours. At the end of the 4-hour period, and with the temperature maintained at 71°C (160°F), the indicator shall be subjected to and shall meet the requirements specified in the turn indicator sensitivity (high temperature) test. There shall be no slip indicator leakage as a result of these tests.

4.6.21 Turn indicator sensitivity (high temperature). The indicator shall be mounted in the normal operating position on a rate table and operated for 10 minutes maximum. The indicator shall then be rotated about the vertical axis at 180 degrees per minute for MS28041-1, MS28041-1A, MS28041-1B, MS28041-1C, MS28041-1D, MS28024-2, MS28024-2A, MS28024-3, MS28024-3A, and 360 degrees per minute for MS28041-2 and MS28041-2A. The pointer deflection shall be 0.31 ± 0.06 inch. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 0.03 of an inch. The test shall be performed in both directions of rotation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

4.6.22 Temperature-altitude. The indicator shall be tested in accordance with Method 504.1, Procedure I, Equipment Category 6 of MIL-STD-810.

4.6.23 Current. A DC ammeter shall be connected in the 28V DC line between the indicator and the source of power. The current consumption at starting

shall not exceed 1.1 amperes. After a period of 3 minutes running time, the current required shall not exceed 0.25 ampere.

4.6.24 Voltage variation. The indicator shall be subjected to and shall satisfactorily pass the turn indicator sensitivity test when operated on an electrical potential of 26V DC and on 30V DC.

4.6.25 Slip indicator leak. The size of the air bubbles in the tube shall be determined with the indicator at room temperature. The indicator shall then be maintained at a temperature of 71° ± 2°C for a period of not less than 1 hour. The size of the air bubble shall then be determined with the indicator at room temperature. There shall be no appreciable change in the size of the air bubble before and after the indicator has been subjected to the high temperature. This test may be combined with the test specified in 4.6.21.

4.6.26 Power requirements. Power requirement shall be for category B equipment. The DC transient levels for transient susceptibility test shall be as follows:

VOLTAGE
TRANSIENT ABNORMAL STEADY STATE LIMIT

(1) With 30 volts applied to the indicator, apply a transient up to the 80 volt level for 50 millisecond.
(2) Repeat the transient pulses 15 times.
(3) Apply a transient up to the 73 volt level for 200 millisecond.
(4) Repeat this 200 millisecond pulse 1 times.
(5) With 28.5 volts applied to the indicator apply 15 pulses (-6 volts). Hold the last 22.5 level for 35 seconds.

NOTE: For the preceding ASSL testing, use a 1 to 5 second interval between pulses.

EMERGENCY STEADY STATE LIMIT

(1) With 22.5 volts applied to the indicator, turn off the power for 7 seconds and then back to 22.5 volts for 10 seconds. Repeat 15 times.
(2) With 28.5 volts applied to the indicator, drop the voltage to 16.5 volts for 2 minutes and then increase to 28.5 volts. Repeat 15 times.

SPIKE VOLTAGE NORMAL STEADY STATE LIMITS



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

(1) With 28.5 volts applied to the indicator, apply a spike up to the +600 volt level for 10 microseconds. Apply 25 of these spikes 1 to 2 seconds apart.

(2) Repeat the above procedure with -600 volt pulses

(3) With 28.5 volts applied to indicator, apply a +160 volt pulse for 100 microseconds. Repeat 25 times.

(4) With 28.5 volts applied to the indicator, apply a -120 volt pulse for 100 microseconds. Repeat 25 times.

NOTE: During this Transient Susceptibility testing, the indicator will be turned about the vertical axis at 180° per minute to check for normal pointer deflection. Paragraphs 4.2.1, 4.2.2 and 4.2.3 (Emergency Operations) of MIL-STD-704 shall apply during these tests. Between pulses pointer deflection should be 0.31 inches ± 0.06 inches. This applies to all transients tests except at the 22.5 volt level, deflection may be 0.31 inches ± 0.12 inches. The indicator current consumption will be monitored during all of the tests and any change in current exceeding 10% is considered a failure. Loss of pointer deflection or abnormal deflection is considered a failure.

4.6.27 Reliability assurance tests. Reliability assurance tests shall be conducted as required by MIL-STD-781. Indicators selected for reliability assurance tests shall first have passed the individual tests.

4.6.27.1 Reliability procedures. Paragraphs titled First article (preproduction) reliability qualification test and production reliability acceptance test (sampling) of MIL-STD-781 shall be applicable.

4.6.27.2 Test level. The equipment inspection conditions shall be of the most severe for each parameter in Table I of those under the Fighter, Transport or Helicopter categories. It shall be established that the indicators under test operate at room temperature. Each indicator shall then be subjected to a low room temperature. The low temperature test shall be conducted in accordance with 4.6.17 and the high temperature test in accordance with 4.6.20. If a particular indicator has already been subjected to tests of 4.6.17 and 4.6.20, the tests need not be repeated but each indicator to be tested under 4.6.27.1 must already have passed the tests of 4.6.17 and 4.6.20. The indicators shall be subjected to the same angular motions as used in the life tests throughout the reliability tests.

4.6.27.3 Duty cycle. The indicator shall operate for a period of 2.5 hours and be off for a period of 0.5 hour.

4.6.27.4 Accept-Reject criteria. Test data shall be evaluated as described in the paragraph titled (Evaluation criteria) of MIL-STD-781.

4.6.27.5 Performance characteristics to be measured. At least once each week the indicator shall be subjected to and meet the requirements of the following tests:



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

Turn indicator static balance (4.6.4)
Turn indicator dynamic balance (4.6.5)
Turn indicator sensitivity (4.6.6)
Slip indicator zero position (4.6.10)
Slip indicator friction (4.6.11)
Slip indicator sensitivity (4.6.13)

A daily observation (on each workday) shall be made of each indicator in order to detect obvious failures.

4.6.27.6 Failure criteria. In addition to the requirements of MIL-STD-781 the following requirements shall be used to determine when a failure has occurred during the test.

Whenever performance characteristics fall below the acceptance requirement (4.6.27.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other to fail.

4.6.27.7 Preventive maintenance. During the period of the tests no preventive maintenance measures may be performed upon the indicator.

4.6.27.8 Length of tests. Each indicator under test shall be operated until either a failure has occurred or an accept-reject decision has been reached. In the event an indicator fails during the test, it shall either be returned to a serviceable condition and again be placed on test or a replacement test indicator shall be placed on test in order that the test sample size shall remain constant throughout the test.

4.6.27.9 Disposition of equipment. Indicators that have been subjected to the reliability tests shall not be delivered on contract until they have been refurbished, resubmitted and passed all the individual tests.

4.6.28 Low temperature exposure. The indicator shall be tested in accordance with Method 502.1, Procedure I, of MIL-STD-810. At the end of the exposure period, the indicator shall be subjected and shall meet the requirements of the turn indicator sensitivity (low temperature) test. The indicator shall then be returned to room temperature and allowed to saturate for a minimum of 4 hours. It shall then be subjected to and shall meet the requirements specified in the individual tests.

4.6.29 High temperature exposure. The indicator shall be tested in accordance with Method 501.1, Procedure I, of MIL-STD-810. At the end of the exposure period, the indicator shall be subjected and shall meet the requirements of the turn indicator sensitivity (high temperature) test. The indicator shall then be returned to room temperature and allowed to saturate for a minimum of 4 hours. It shall then be subjected to and shall meet the requirements specified in the individual tests.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

4.6.30 Acceleration: Procedure I. The indicator, not operating, shall be mounted and accelerated, first in its vertical axis, and then in each of two axis that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g's in each of the above axis for a period of 30 seconds. At the end of this acceleration, the indicator shall meet the requirements specified for turn indicator sensitivity. No damage to the indicator shall result from the test.

4.6.31 Salt fog. The indicator, with the gyro not operating but with the external connections made to simulate installed conditions, shall be tested in accordance with Method 509.1 of MIL-STD-810. At the conclusion of the test, the indicator shall be subjected to and shall meet the requirements specified in the following tests.

Low voltage operation
Turn indicator static
Turn indicator dynamic balance
Turn indicator sensitivity
Turn indicator damping

4.6.32 Mounting lugs. The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for 1 minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture, and there shall be no damage to any part of the indicator. The indicator shall then be subjected to and shall meet the requirements specified in the seal test.

4.6.33 Temperature shock. The indicator shall be tested in accordance with Method 503.1 of MIL-STD-810. At the end of this test, the indicator shall be subjected to and shall meet the requirements of the seal test.

4.6.34 Radio noise interference. The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with MIL-STD-461 and MIL-STD-462.

4.6.35 Test for helium. The indicator case shall be punctured, or the filling tube cut, and the indicator subjected to the test for helium by means of mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.36 Internal examination. The case of the indicator shall be opened and the internal mechanism examined. Any deterioration or damage which could in any manner prevent the indicator from meeting functional operation or maintenance requirements during service life shall be cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

4.7 Presubmission testing. No items, part, or complete indicator shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and retest. Indicators which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

4.9 Inspection and test. Test of Methods of Preservation and Packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

4.10 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspeciton of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (6.2).

5. PACKAGING:

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (see 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IC-1 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

5.2.1 Container-Cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall be of sufficient density and thickness to protect the unit container from punctures. Intermediate container requirements shall conform to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.3. Level B. Same as Level "A" unless otherwise specified by special packaging instructions (SPI) or the contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, special packaging instructions (SPI) and MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The turn and slip indicator is intended for use in aircraft to indicate the rates of rotation of the aircraft about its vertical axis and the lateral attitude of the aircraft relative to the apparent vertical.

6.2 Test values. Normal and limiting values of performance data should be determined at input voltages of 27.5 ± 0.5V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

6.3 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance an reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.4 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts of tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a MTBF greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence will apply:

(a) Contract: The contract will have precedence over any specification.

(b) This specification: This specification will have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, must be specifically approved in writing by the procuring activity.

(c) Referenced specifications: Any referenced specification will have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications will apply to the extent specified.

6.6 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) The quantity, and MS part number of the instrument desired.
(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.
(d) Method of preservation, if other than Method 1C1 (see 5.1).
(e) Whether government tests are required and if required, the laboratory at which government tests are to be conducted.
(f) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

6 6 1 Contract data requirements Data conforming to Data Item Descriptions DI-T-3709/T-109-2 and DI-T-3718/T-119-2 will usually be required for delivery in connection with this specification When so required, such data will be specified for delivery on a DD Form 1423 included in the contract (see 4 4 2)

6 7 Definition of hermetic seal A hermetic seal is defined as a perfectly closed and airtight seal made between metallic and metallic or between metallic and vetric materials A hermetic seal is not intended to include seals accomplished by gaskets

6 8 Subject term (key word) listing

Hermetically
Turn and Slip Indicator
28 VDC

6 9 International standardization Certain provisions of this document are subject to international standardization agreement ASCC Air Std 10/5 and STANAG 3322 When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations

6 10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodians
Army -AV
Navy -AS
Air Force -99

Preparing Activity:
AIR FORCE -71

Project Number
6610-0362



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1 DOCUMENT NUMBER | 2 DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-I-7627F | 15 NOVEMBER 1994 |

3 DOCUMENT TITLE

INDICATOR, TURN AND SLIP

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (*Include Zip Code*) | d. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br>(2) AUTOVON (*If applicable*) | 7. DATE SUBMITTED (*YYMMDD*) |

8 PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (*Include Area Code*) |
|---|---|
| OC-ALC/TICLA | (1) Commercial (2) AUTOVON 336-5960 |
| c. ADDRESS (*Include Zip Code*)<br>3001 STAFF DR.<br>TINKER, AFB, OK 73145-5990 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041 3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

DD Form 1426, OCT 89 — Previous editions are obsolete. — 198 290

Source: https://assist.dla.mil - Downloaded: 2026-10-17T08:40Z
Check the source to verify that this is the current version before use.